

- (11 pt) 1. Use the provided phylogeny of living fishes to fill in the blanks below with the appropriate systematic group name and classification level (class, subclass, etc.). Each lettered blank below corresponds to the lettered branch on the phylogeny. (1/2 point per blank). Spelling counts throughout.

	Group Name	Classification Level
A.	_____	_____
B.	_____	_____
C.	_____	_____
D.	_____	_____
E.	_____	_____
F.	_____	_____
G.	_____	_____
H.	_____	_____
I.	_____	_____
J.	_____	_____
K.	_____	_____

- (1 pt) 2. Groups J and K together form what group and classification level?

- (1 pt) 3. Groups C and D together form what group and classification level?

- (1 pt) 4. Which group or groups form the Actinopterygii? Which classification level is this?

- (1 pt) 5. Which group or groups form the Neopterygii? Which classification level is this?

Match the *most appropriate* term to each concept. No term is used twice. (2 points each)

Ampullary	Cypriniformes	Otoliths	Schreckstoff
Astericus	Diphycercal	Paired species	Sensory Hair Cell
Branchiostegal Ray	Electrolary	Pelvic claspers	Serraniformes
Canal Neuromast	Gular Plate	Perciformes	Siluriformes
Cartilaginous Skeleton	Kinocilium	Piranhiiformes	Stereocilium
Catistomiformes	Lapillus	Pussenboots	Superficial Neuromast
Ceratotrichia	Lepidotrichia	Sagitta	Tuberos
Cupula	Lorenzini	Schreckendonkey	Weberian Ossicles

6. _____ Largest order of freshwater fishes.
7. _____ Structure found only in Otophysan fishes.
8. _____ Name any one order of Otophysan fishes, except your answer from question 6. (Bonus +1 if you use an order not in list above.)
9. _____ Electrical receptor used for intraspecific communication.
10. _____ Diagnostic feature of Chondrichthyes.
11. _____ Sensory structure found inside the lateral line.
12. _____ Substance released by some fishes when bitten.
13. _____ One species of lamprey is a micropredator. Its sister species is not.
14. _____ Specific otolith used for orientation.
15. _____ Name any family from the Perciformes that is found in Missouri. The name is not in the list above!

(GSH) Answer the next two questions. Terms are not in the word bank. (2.5 points each.)

16. _____ Type of scales found on Cladistia.
17. _____ Superclass with two semi-circular canals.

Examine the photo on screen, and then answer the next three questions.

- (1 pt) 18. To what class does the fish on screen belong? (Hint: not ichthyology class.)
- (4 pt) 19. List two diagnostic features that you used to determine your answer to the previous question.
- (5 pt) 20. Describe a mechanism that the fish on screen can use to reduce the chance of being killed by predation. Be sure to explain how the mechanism you describe reduces the effectiveness of predators.
- (10 pt) 21. Describe the conditions that favor cockholdry in *Lepomis* sunfishes. Name and describe both types of strategies used by young males.

(15 pt) 22. List and briefly explain the difference between the two types of diadromy related to reproduction. Explain any two adaptive advantages of diadromy. Name a representative group of fishes that displays each type. You do not have to use the scientific names for the groups.

(5 pt) 23. **(GSH)** Name the hemisphere (northern, southern) and latitude (high, low) most commonly associated with each type of diadromy you named above.

(1 pt) 24. **Extra Credit:** Name either of the two largest families of perciform fishes in the world.

- (15 pt) 25. Name, compare and contrast the three types of jaw suspension found in fishes. Indicate which type of suspension is most commonly found in actinopterygian fishes. For the other two types of jaw suspension, name one major group of fishes that has that type. Do not repeat group names.

- (15 pt) 26. Describe the observed evolutionary trends of the following structures in the Actinopterygii: scales, swim bladder, caudal fin, and paired fins. Clearly explain how / why the evolutionary changes of these four structures are interrelated. That is, explain how / why the change in one particular structure reinforced the evolutionary changes of the other structures.